

Chapter 8 — Brain and model

The question from Chapter 1, answered

Is this like a brain?

Seven chapters ago this was premature. You have now priced the thing and watched it fail, so the answer is worth something.

Start with what is not true

- Transformers were not derived from neuroscience. They came out of machine translation, and the design pressures were throughput and parallelism.
- "Attention" shares a name with neural attention and little else. Chapter 1 gave you the definition: a weighted sum over key–value pairs, with weights from a compatibility function. That is linear algebra with a borrowed noun.
- The resemblance people feel is mostly that both produce language. That is the thing being explained, not evidence for the explanation.

What *is* defensible

There is a real literature here, and it is more careful than its summaries.

- Take a language model, feed it the same text a person read or heard, and extract the model's internal activations.
- Fit a regression from those activations to the person's brain responses.
- Ask how much of the response you can predict. The answer is: a surprising amount.

Say the method correctly. This is an **encoding model** — ridge regression from layer activations. It is *not* "next-token surprisal predicting neural responses", which is how the result usually circulates, including in an earlier draft of this course's own notes.

Scope: 43 models, against three neural datasets of $n = 10$, $n = 5$ and $n = 5$. And the *predictive-processing* conclusion is not measured within a brain at all — it is a **between-model correlation**: models that predict the next word better tend to score better on brain data. Hold onto that, because it is precisely what gets attacked two slides from now.

The headline needs its denominator

The result is often quoted as predicting nearly 100% of explainable variance.

- That figure is normalised by a **noise ceiling** — how much of the signal is predictable even in principle, given how noisy the recordings are.
- The ceilings in that work are 0.32, 0.17 and 0.20 across the three datasets.
- The authors say so themselves, and compare unfavourably: *"even this ceiling is low relative to single cell recordings in the primate ventral stream e.g., 0.82 for IT recordings."*

Be precise about what a **ceiling** is. It estimates the reliability of the *recordings* — how much of the signal is predictable in principle given repeat-to-repeat noise. It is a fact about the dataset, not about the size of the brain–model correspondence.

- So it licenses this: raw predictivity is low in absolute terms, the normalised headline is doing a lot of work, and the denominator is itself an estimate with error attached.
- It also licenses the authors' own point — these datasets **separate models far less sharply** than primate recordings do.

And the result that complicates it

- In the same work, untrained models with a trained linear readout performed well above chance.
- If a randomly initialised network predicts brain data respectably, then what is being measured is not straightforwardly "what training taught it".
- The authors report this. It is in the paper, not in the summaries.
- **Hold this one loosely.** Two slides from now it stops being a curiosity and becomes a confound.

Worth naming for a course about citation discipline: this paper went through the PNAS **Contributed** track, where the contributing author selects the reviewers. That is not a criticism of the work. It is a fact about the review process, and you would want to know it about your own field.

The strongest version, and its own authors' limits

A second line of work records neural activity directly, with electrodes, while people listen to speech.

- Scope, which matters: **nine participants, all epilepsy patients with implanted electrodes, listening to one thirty-minute podcast**, compared against GPT-2.
- The shared structure they find is real and it is specific.

The authors then say this, and it is the sentence this chapter is built on:

These shared computational principles, however, do not imply that the human brain and DLMs implement these computations in a similar way.

While transformer models are an impressive engineering achievement, **they are not biologically feasible.**

The field correcting itself, in public

- One re-analysis attacks the between-model correlation directly: predicting future words **does not uniquely, or even best, explain** why a representation fits brain data. Within a model, the representations best at prediction are *"strictly worse brain models"* than others.
- Another re-analyses **exactly the three datasets** the first result rests on. Shuffled train–test splits produced spurious results — and **positional signals and word rate fully account for the neural predictivity of untrained models**.
- That second finding **dissolves** the untrained-model result from two slides ago rather than complicating it. It was a confound, not a discovery.

Idan Blank is a co-author of the original paper and of the critique. That is what a healthy field looks like from outside, and it is the cleanest example in this course of the habit Chapter 17 asks of you.

[V] Antonello, R. & Huth, A., 2024. *Neurobiology of Language* 5(1):64–79, DOI 10.1162/nol_a_00087.

[V] Hadidi, N., Fegghi, E., Song, B. H., Blank, I. A. & Kao, J. C., 2026. *Nature Communications* 17(1):5769, DOI 10.1038/s41467-026-72253-7.

So: run the analogy, then break it

The analogy is worth running. It is the break that carries the content.

- The hippocampus consolidates episodic experience into durable memory. Something happens, and it is written into a store that changes you.
- A model at inference does none of this. The weights are frozen. Nothing that happens in a session is retained anywhere.
- The context window is not a hippocampus. It is a buffer that gets discarded.

The comparison everyone makes, and why it cannot be made

You will hear this one: human working memory holds about four items, a context window holds hundreds of thousands of tokens, **so the machine wins on memory.**

- The primary source does not say four. It argues for **three to five chunks**, and only under conditions engineered to block chunking and rehearsal.
- It is a target article published with roughly seventy pages of peer commentary in the same issue, and the fixed-slot view it defends is **actively contested** — competing accounts treat capacity as a shared resource rather than a count of slots.

But the deeper problem is not that the number is wrong. A *chunk* is a unit of meaning assembled by the person recalling it. A *token* is a fragment of text fixed by a vocabulary before training. There is **no conversion between them**, so no ordering between 4 and 200,000 is defined at all.

This is not a weak comparison. It is **not a comparison** — the same failure as dividing by a denominator whose units you never stated.

What you can say instead

Drop the counting and the asymmetry is still there — and it never needed a number.

Context window

- Supplied from outside. You put things in it.
- Passive. Holding a token costs nothing and does nothing; it sits there.
- Contents are **fixed** for the pass and then discarded whole.

Working memory

- Maintained from inside, and it costs effort to hold anything.
- **Active and interference-prone** — items compete, and rehearsal is what keeps them.
- Continuously **rebuilt from and written back into** long-term memory.

One is a **buffer**. The other is a **process inside a system that consolidates**. That difference does not get smaller if you make the buffer larger, which is the whole point — and it is why Chapter 10 hands you a file rather than a bigger window.

One more comparison to refuse

- *"The frontal cortex is the reasoning part, so the later layers must be the reasoning part."*
- The premise is **contested within neuroscience itself**. Borrowing a disputed claim from one field to explain another does not produce understanding; it produces two problems.
- Chapter 5 already told you the related thing: the model's stated reasoning is not an explanation of the model's behaviour.

What the analogy is actually good for

- As a source of hypotheses, it has produced real, checkable work — that is the literature above.
- As an explanation, it does not survive its own authors' caveats.
- As a guide to what the tool will do on Thursday, it is worse than useless, because it imports intuitions about memory, understanding and intent that the system does not have.

You have a better guide. It is Chapters 1 through 7.

Where Session 1 ends

- A next-token predictor, shaped by a rating process you have now occupied.
- Measurable, and mostly weakly, on five cases.
- Failing in ways that are structural rather than accidental, and expensive in ways that have an equation.
- Not a brain — and the interesting part is *which* difference matters.

The model has no hippocampus. **So you have to be its hippocampus.**

That is Session 2, and it starts with a system that acts.